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CATCHWORD

Inherently Interpretable Machine Learning: A Contrasting Paradigm to Post-hoc Explainable AI

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1 Introduction

A decade ago, data scientists devoted roughly 60–80% of their effort to manual feature engineering, aiming to create features with high predictive power and meaningful properties (Press 2016). However, with the rise of deep learning and highly flexible models capable of processing raw input features and learning intricate feature representations, this focus has shifted dramatically (Janiesch et al. 2021). Today, data scientists spend much of their time explaining opaque machine learning (ML) models because users struggle to understand their complex decision logic (Bauer et al. 2021).

As ML models have become integral to data-driven decision-making across many disciplines, including Business and Information Systems Engineering (BISE) and Information Systems (IS) (Abdel-Karim et al. 2021; Duda et al. 2024), the need for model comprehensibility is more critical than ever (Bauer et al. 2021; Brasse et al. 2023). That is, human users must be able to understand the process of how an ML model comes from input data to its predictions based on its internal structure and the relationships it has learned. This property is critical not only for building confidence in ML models for high-stakes domains like healthcare and finance, but also for debugging flawed model structures, supporting regulatory and ethical compliance, enabling users to learn from ML insights, and fostering broader and more responsible adoption in business environments.

To address the challenge of lacking model comprehensibility, the BISE/IS community has primarily focused on the use of post-hoc explainable approaches from the field of explainable artificial intelligence (XAI) (e.g., Brasse et al. 2023; Kim et al. 2023; Kruschel et al. 2025). This paradigm emphasizes using highly flexible black-box models that achieve high predictive performance at the expense of comprehensibility. As a result, such models require supplementary explanation techniques, such as Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al. 2016), SHapley Additive exPlanations (SHAP) (Lundberg et al. 2020), or RObust Local EXplanations (ROLEX) (Kim et al. 2023), to translate their complex decision logic into forms understandable by humans (Bauer et al. 2021). While effective, this paradigm has several limitations, including potential inaccuracies and oversimplifications in the generated explanations (Rudin 2019).

In contrast to the reliance on post-hoc explainability, there is an alternative paradigm that focuses on inherently interpretable ML models. Unlike black-box models, these

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models are designed to be intrinsically comprehensible, allowing stakeholders to understand their structure and decision-making processes without the need for additional explanation techniques. This is achieved through carefully crafted model constraints that ensure that the model remains both interpretable and effective (Kruschel et al. 2025; Rudin 2019). As a result, the paradigm balances the needs for feature engineering and additional model explanation. However, despite some clear benefits of inherently interpretable ML models, especially in the context of interpretable input features, they have not yet been widely adopted in the BISE/IS community, which often remains focused on black-box models and post-hoc explainable techniques (e.g., Brasse et al. 2023; Schoormann et al. 2023).

A critical issue in this area is the widespread confusion or disagreement within the BISE/IS community regarding these two distinct paradigms. Inherently interpretable models and post-hoc explainable approaches are sometimes treated as synonymous, leading to potential misunderstandings regarding their fundamental differences (Longo et al. 2024). Furthermore, the misconception persists that only flexible black-box models are capable of achieving high predictive performance across all types of data, ignoring the fact that inherently interpretable models deliver competitive results in specific contexts. This calls for a more differentiated view that considers the nature of the underlying data. While complex black-box models typically demonstrate advantages in settings with unstructured, high-dimensional data, such as text and image, inherently interpretable models have proven highly effective for structured, tabular data in terms of both predictive performance and comprehensibility (e.g., Rudin 2019; Kruschel et al. 2025). Another misconception is the belief that inherently interpretable models are limited to overly simplistic approaches, such as linear regression (e.g., Bauer et al. 2021; Kim et al. 2023). This narrow view overlooks the growing body of research, demonstrating that inherently interpretable models can be highly sophisticated and capable of capturing intricate patterns in data while remaining comprehensible. Advanced approaches, such as modern variants of generalized additive models (e.g., Kraus et al. 2024), optimal decision tree models (e.g., Hu et al. 2019), and subscale models (e.g., Chen et al. 2018), have significantly expanded the potential of inherently interpretable models, allowing them to offer a robust alternative to black-box models, without sacrificing too much predictive performance.

This catchword article aims to address these misconceptions and provides clarity on the conceptual distinctions between inherently interpretable models and post-hoc explainable approaches. We argue that the community must engage in a more nuanced reflection on the strengths, limitations, and use cases of both paradigms. Without a deeper understanding of these issues, there is a risk that the field may overemphasize black-box models without fully considering the benefits of inherently interpretable alternatives. Moreover, we argue that the BISE/IS community, as a socio-technical and design-oriented discipline, has great potential to drive significant advancements in the field of inherently interpretable machine learning (IIML) by combining algorithmic innovations with user-centric needs to develop new models, methods, and information systems for more effective decision support.

2 Foundations of Inherently Interpretable Machine Learning

2.1 Conceptual Foundations and Distinction Between Interpretability and Explainability

The term 'interpretable machine learning' has been the subject of extensive conceptual debates across various disciplines (e.g., Broniatowski 2021; Lipton 2018; Miller 2019). At the heart of these debates lies the concept of interpretability, which is closely related to adjacent notions, such as comprehensibility, transparency, and explainability. This semantic similarity has led to a terminological ambiguity, risking confusion even among seasoned researchers and practitioners (Longo et al. 2024). Indeed, a closer look at exemplary definitions provided in Online Appendix A (available online via http://link.springer.com) reveals that there is no universally accepted understanding of these terms across disciplines, and perhaps such a consensus is neither feasible nor beneficial (Longo et al. 2024). Nevertheless, certain terminological tendencies have emerged that can offer useful guidance, particularly for the interdisciplinary BISE/IS community (see Fig. 1).

On a more general level, it can be said that, in the context of AI and ML models, *interpretability generally refers to a model's ability to explain or present its behavior in human-understandable terms* (Doshi-Velez and Kim 2017; Guidotti et al. 2018). This ability is often associated with attributes such as model comprehensibility and transparency, referring to transparent model properties that can clearly demonstrate how input features contribute to a prediction in an intuitive, human-comprehensible way (e.g., additive combination of features in a linear regression or partial dependence plots visualizing feature-target relationships) (Barredo Arrieta et al. 2020; Lipton 2018).

Moreover, the term interpretability intersects with the broader concept of explainability, particularly as part of the widely used umbrella term 'explainable artificial intelligence', which has gained increasing prominence since 2015 (e.g., Barredo Arrieta et al. 2020; Gunning et al.

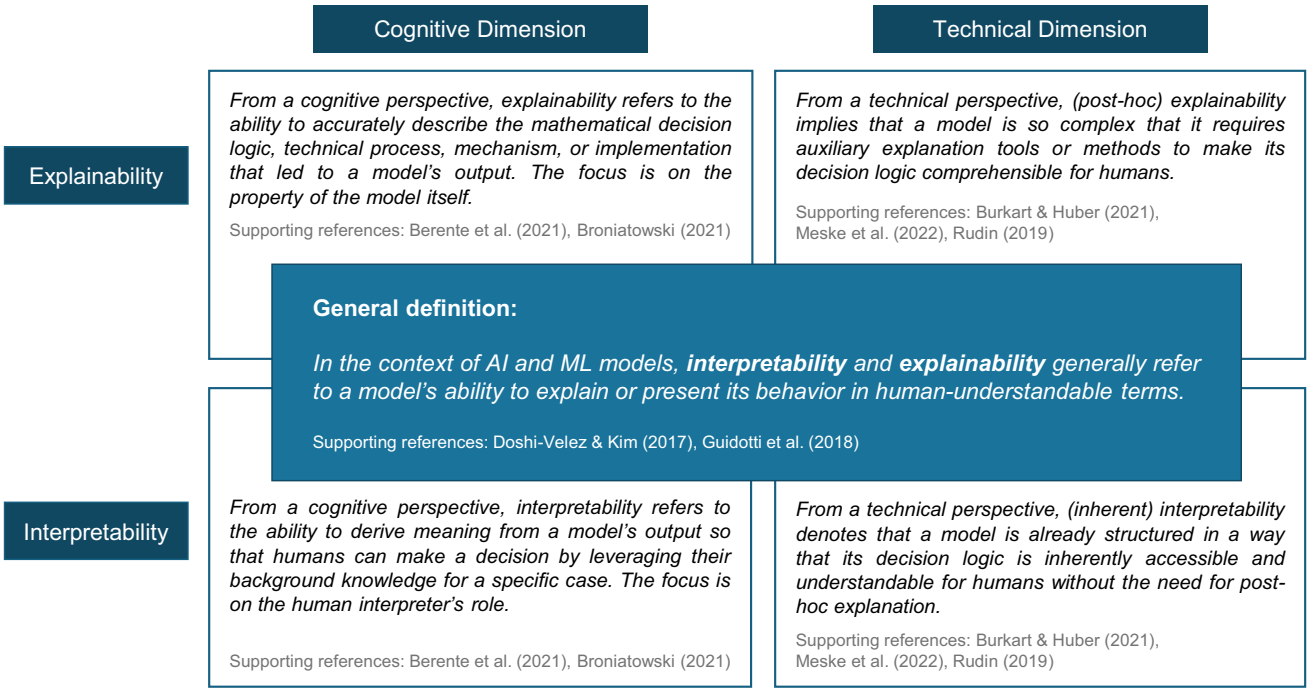


Fig. 1 Terminological dimensions and differences for the terms 'interpretability' and 'explainability'

2021; Longo et al. 2024). While both terms can be treated as synonyms when used as overarching concepts (e.g., Bauer et al. 2021; Du et al. 2019; Miller 2019), the deliberate choice between the terms 'interpretability' and 'explainability' may also signal important distinctions. These distinctions can be characterized along two dimensions: a cognitive dimension and a technical dimension.

From a cognitive-psychological perspective, interpretability can be seen as the ability to derive meaning from a model's output so that humans can make a decision by leveraging their background knowledge for a specific case. In contrast, explainability can be seen as the ability to accurately describe the mathematical decision logic, technical process, mechanism, or implementation that led to a model's output (Broniatowski 2021). In this view, explainability rather emphasizes the properties of the model itself, whereas interpretability emphasizes the human interpreter's role in making sense of the information (Berente et al. 2021). However, the process of making sense of information is highly subjective and context-specific. For example, a medical expert may require detailed feature-level insights to verify a model's recommendation in a diagnostic setting, while a financial regulator may be more concerned with the justification of credit decisions based on straightforward rules. This variability makes it difficult to establish universal criteria for what constitutes interpretability from a cognitive perspective. We discuss this issue in more detail in Sect. 4.1, outlining central challenges and opportunities for future research.

From a technical perspective, the distinction between interpretability and explainability is more straightforward and mainly stems from the complexity of the underlying ML model, resulting in two distinct paradigms with different needs for additional explanation. That is, explainability implies that a model is so complex that it requires auxiliary explanation tools or methods to make its decision logic comprehensible for humans. In contrast, interpretability denotes that a model is already structured in a way that its decision logic is inherently accessible and understandable for humans without the need for post-hoc explanation (Burkart and Huber 2021; Meske et al. 2022; Rudin 2019). To clearly differentiate between these two paradigms, it is advisable to apply unambiguous qualifiers such as 'post-hoc' versus 'inherent' to the respective terms (e.g., post-hoc explainability vs. inherent interpretability) (Du et al. 2019; Kraus et al. 2024). Without clearly distinguishing between these two paradigms, there is a risk of conceptual confusion, which could result in significant advancements being overlooked. Recent reviews in the BISE/IS literature, for example, have focused exclusively on post-hoc explainable approaches (e.g., Brasse et al. 2023), ignoring many inherently interpretable models that have existed well before the rise of XAI (e.g., Freitas 2014). Against this backdrop, this catchword article aims to clearly distinguish between both paradigms to avoid a one-sided perspective. This is important because the unreflected use of post-hoc explainable methods in critical areas can lead to unwanted consequences (see Sect. 3).

Furthermore, it should be noted that it is tempting to assume that ML model explanations, whether post-hoc or inherent, can be used to explain real-world phenomena in a causal manner. Although the discovery of causal insights is among the most desired goals when constructing data-driven models (Barredo Arrieta et al. 2020; Miller 2019; Longo et al. 2024), off-the-shelf models and methods in the realm of IIML and XAI usually do not claim to answer causal questions, such as identifying cause-effect relationships or estimating the impact of interventions (Molnar et al. 2022). This is because traditional ML models are designed to learn correlational patterns and exploit statistical associations between variables. Thus, post-hoc explainable or inherently interpretable model explanations can be used to describe how a model processes input data to produce predictions, but they do not disentangle correlation from causation and fail to account for confounding variables, proxy effects, overfitting, or other factors that may distort causal interpretations (Caruana et al. 2015; Longo et al. 2024; Molnar et al. 2022). Without a formal causal framework, correlation-based model explanations risk being mistaken for causal evidence, potentially leading to misguided conclusions and actions. Recognizing this limitation is particularly important in scientific and organizational contexts, where the objective is often to gain a genuine understanding of the true data-generating process, rather than just making predictions. Nevertheless, techniques from both paradigms can serve as valuable exploratory tools to guide subsequent causal analyses and support theory development and testing. We discuss this aspect in more detail in Sect. 4.6.

2.2 Mathematical Notations

Mathematically, an inherently interpretable model $\mathcal{M} \in \mathcal{H} : \mathcal{X} \rightarrow \mathcal{Y}$ maps feature vectors $x \in \mathcal{X}$ to a response variable $y \in \mathcal{Y}$, with $\mathcal{H}$ denoting the hypothesis space. The objective is to train the model $\mathcal{M}$, by minimizing the empirical risk

$$\min_{\mathcal{M} \in \mathcal{H}} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\left(y^{(i)}, \mathcal{M}(x^{(i)})\right), \tag{1}$$

where N is the number of instances in the training dataset, $x^{(i)} \in \mathcal{X}$ and $y^{(i)} \in \mathcal{Y}$ are the i-th data instance and response, respectively, and $\mathcal{L}$ is a loss function that measures the error between the true response variable $y^{(i)}$ and the model's prediction $\mathcal{M}(x^{(i)})$.[1] The idea of IIML is to constrain the hypothesis space $\mathcal{H}$ so that only ML models can be found that are easily understandable to humans. To achieve this, the hypothesis space can be constrained using domain-specific structural knowledge (e.g., monotonicity) or physically-informed constraints, which enhance the comprehensibility of the model's structure and behavior (Rudin 2019).

Insights from IIML models can generally be derived in the form of feature importance, i.e., the undirected relevance of a feature for predicting a response variable; or feature effects, i.e., the information about how the model prediction changes for varying values of a feature. For example, in a decision tree, the importance of a feature is determined by how often and how effectively the feature is used to split data instances throughout the entire tree. In contrast, in a generalized additive model (GAM), the effect of a feature is determined based on a so-called shape function. It is important to note that both feature importance and feature effect are consistent characteristics of IIML models and independent of any samples.

Depending on the structure of the IIML model, constraints may be relaxed to allow modeling of more complex relationships between the features and the response variable. For example, for GAMs, which typically do not consider interactions between features, it may be possible to include interaction effects. An interaction between two features is called a pairwise interaction, whereas an interaction between more than two features is called a higher-order interaction. However, because higher-order interactions are inherently complex and difficult for humans to understand, IIML models that capture such interactions are typically considered less interpretable (e.g., Yang et al. 2021).

2.3 Exemplary Inherently Interpretable Models

Among the many representative models of IIML, a few candidates have become particularly prominent because they have shown consistently favorable results in a variety of ML tasks (e.g., Kruschel et al. 2025; Rudin 2019; Wang et al. 2023). Table 1 provides an overview of these models, including advantages and limitations. Thereby, most IIML models are designed to process interpretable input features. For unstructured data, such as time series or images, additional preprocessing steps are required (see Sect. 3.3), but initial research also aims to extend IIML model architectures to be able to process raw unstructured data in its natural form (e.g., Zilker et al. 2024).

To illustrate the structure and inner workings of these IIML models, we consider the case of predicting diabetes given medical and demographic data from patients (Age, body mass index (BMI), and HbA1c (blood glucose)

[1] Please note that in this article, we explicitly focus on supervised learning as one of the most important subfields of IIML. For further considerations on unsupervised learning and reinforcement learning, please refer to Allen et al. (2023) and Glanois et al. (2024).

Table 1 Overview of different types of inherently interpretable machine learning models

Model	Brief explanation	Pros	Cons	Examples
Scoring models	Scoring models use a set of rules to assign points to features, which are then summed to produce a prediction. These models are highly interpretable, as each rule directly shows the impact on the prediction. They handle interactions between predictors effectively, but can become unwieldy with large rule sets	Simple, interpretable, and transparent logic Captures interactions between features Requires little to no preprocessing	Large rule sets can reduce interpretability Overfitting risk without proper constraints	Ustun and Rudin (2019)
Optimal decision trees	Learned optimal decision trees oftentimes learn simpler trees by being optimized globally instead of greedily. Trees are intuitive, easy to visualize, and handle non-linear relationships effectively. However, small changes in data can drastically alter the tree structure	Easy to interpret and visualize with intuitive splits Handles non-linear and categorical data well Requires minimal preprocessing (no scaling needed)	Overfitting risk without pruning or constraints High sensitivity to small data changes Lower predictive performance compared to advanced models	Bertsimas and Dunn (2017), Hu et al. (2019)
Generalized additive models (GAMs)	GAMs use shape functions to model non-linear relationships while maintaining interpretability. The effect of each feature is represented by its own shape function, assuming additive effects, which allows easy visualization but limits the ability to model complex higher-order interactions	Captures non-linear patterns while staying interpretable Flexible smoothing for individual features	Assumes additive effects, missing complex interactions More computationally intensive than simple models	Caruana et al. (2015), Yang et al. (2021), Kraus et al. (2024)
Hybrid models	Hybrid models integrate multiple approaches (e.g., linear models with decision trees) to improve predictive performance while maintaining interpretability. They balance the strengths of different models but may require additional effort in tuning and interpreting the combined outputs	Captures complex, non-linear interactions Balances interpretability and flexibility Combines strengths of different models	Interpretation may decrease with complexity Tuning combined models is more challenging	De Caigny et al. (2018), De Bock and De Caigny (2021)
Subscale models	Subscale models divide features into meaningful groups, modeling each group separately before combining results. This approach improves interpretability by focusing on specific feature sets. Feature grouping requires careful domain expertise to avoid introducing bias	Provides interpretable insights into feature groups Simplifies analysis by reducing model complexity	May miss global interactions across groups Requires domain knowledge for feature grouping	Chen et al. (2018)

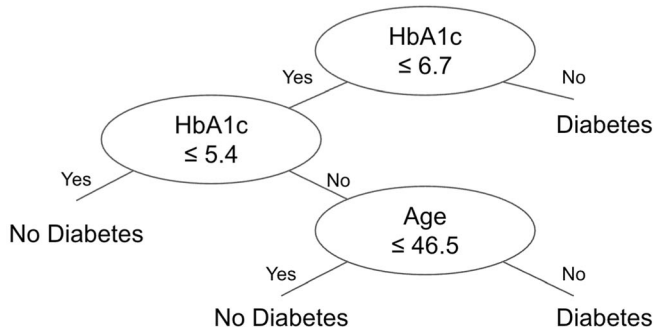
level).[2] We perform minimal preprocessing of the data by simply standardizing the features.

From a methodological perspective, IIML models can generally be classified based on the trade-off between interpretability and complexity. Highly interpretable models prioritize simplicity and transparency, making them easy to understand for both domain experts and non-technical users. Examples include scoring models and optimal decision trees. Scoring models use straightforward additive rules to assign points to features, producing predictions that are easy to trace and explain. Such models have been successfully applied in risk scoring systems due to their transparency and simplicity (e.g., Ustun and Rudin 2019). As shown in Fig. 2a, in the case of predicting diabetes, we find easily interpretable rules, such as an increased risk of 0.11 if a patient is older than 37 and has an HbA1c value greater than 5.4 To make a final prediction, the risks (rows in the figure) are simply summed up. If the value exceeds a certain threshold, the prediction is 1 (prediction of diabetes), else 0 (prediction of no diabetes). Similarly, optimal decision trees provide intuitive, hierarchical splits that are inherently interpretable while capturing non-linear relationships (e.g., Bertsimas and Dunn 2017; Hu et al. 2019). As illustrated in Fig. 2b, optimal decision trees produce a similar outcome to traditional ones. However, optimal decision trees oftentimes learn simpler trees by being optimized globally instead of greedily. Although highly interpretable and, thus, applicable also in high-risk cases,

[2] We use a publicly available dataset from https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset (accessed 15 August 2025).

Rules	Coefficient
Age > 37.0 AND HbA1c > 5.4	+0.11
Age > 38.5 AND HbA1c > 5.4	+0.17
Age > 43.5 AND HbA1c > 5.4	+0.15
BMI ≤ 29.7 AND HbA1c ≤ 6.7	-0.25
Age ≤ 52.5 AND HbA1c ≤ 6.7	-0.19

a)

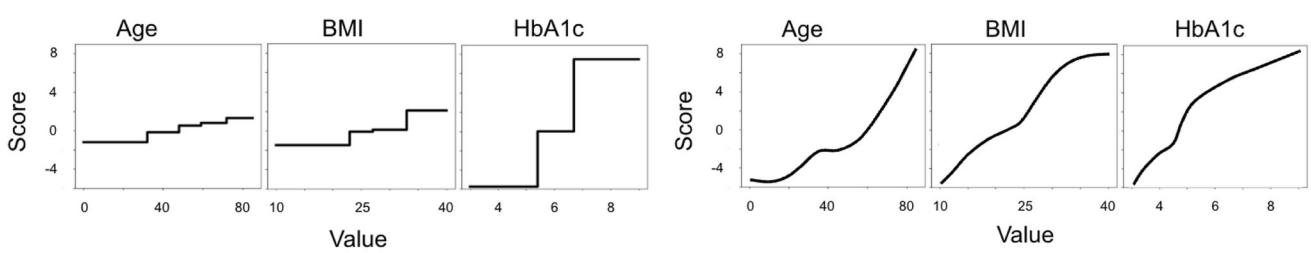


b)

Fig. 2 Simple representatives of IIML models trained on the task of predicting diabetes based on features Age, body mass index (BMI), and HbA1c (glucose level). **a** Shows a scoring model, and **b** shows an optimal decision tree

these simple models may struggle with more complex patterns in data or require constraints to mitigate issues like overfitting.

IIML models of medium complexity aim to combine interpretability with the flexibility needed to capture non-linear relationships and interactions. GAMs exemplify this category by allowing each feature to be modeled with a so-called shape function while maintaining additive effects, which simplifies visualization and explanation (e.g., Caruana et al. 2015; Yang et al. 2021). However, these models assume additive relationships and may miss interactions unless they are specifically extended, as in Explainable Boosting Machine (EBM) (Caruana et al. 2015). In the case of predicting diabetes, Fig. 3a and b show two representatives of GAMs. Figure 3a shows the output of a trained EBM model, based on tree stumps and, thus, showing the typical step-wise shape functions. Figure 3b shows the output of Interpretable Generalized Additive Neural Networks (IGANN) (Kraus et al. 2024), representing smoother shape functions that potentially improve the interpretability but can lack accuracy when the effect of a feature changes dramatically for a small change in the feature value. Hybrid models, such as the Logit Leaf Model, extend the model flexibility further by integrating tree-based splitting with simple models, enabling them to capture interactions while retaining interpretability (e.g., De Caigny et al. 2018). This balance makes such models particularly attractive for applications requiring nuanced explanations without sacrificing prediction performance. In the case of predicting diabetes, Fig. 3c shows a trained



a) b)

Rules	Intercept	Age	BMI	HbA1c	Notes
HbA1c ≤ 5.4	—	—	—	—	Pure Leaf (No Diabetes)
HbA1c ≤ 6.7 & HbA1c > 5.4 & Age ≤ 43.5	-0.590	+0.054	+0.070	-0.489	
HbA1c ≤ 6.7 & HbA1c > 5.4 & Age > 43.5	-0.932	+0.037	+0.102	-0.718	
HbA1c > 6.7	—	—	—	—	Pure Leaf (Diabetes)

c)

Fig. 3 IIML models of medium complexity trained on the diabetes task. **a** shows the shape functions of an EBM model, and **b** shows the shape functions of an IGANN model, both are representatives of GAMs. **c** shows the Logit Leaf Model, a hybrid model combining a decision tree and multiple logistic regressions

Logit Leaf Model. Specifically, the left column 'Rules' describes how the patients should first be split into different subgroups (e.g., all patients with $5.4 \geq$ HbA1c ≤ 6.7 and Age ≤ 43.5 end up in the second subgroup). Within this subgroup, the linear model is described in the right columns.

At the end of the complexity spectrum, approaches like subscale models focus on capturing nuanced feature effects, often at the cost of increased computational demands or extensive modeling effort. Subscale models divide features into meaningful groups and model each group separately, improving interpretability within specific feature sets while requiring domain expertise to define the groupings (e.g., Chen et al. 2018). These approaches are particularly useful in domains where a detailed understanding of specific feature subsets is critical, such as healthcare or finance. In the case of diabetes prediction, we formed an age group, simply including the feature 'Age', and a metabolic group, comprising the features 'BMI' and 'HbA1c'. For both groups, we trained a logistic regression model to predict diabetes, respectively. The predictions from both logistic regression models then function as inputs to a meta model, another logistic regression model. The meta model shows that both the age group and the metabolic group have a positive effect, with the latter potentially having a larger effect (see Fig. 4).

Furthermore, it should be noted that not only the type of model determines the degree of interpretability, but also the choice of model configuration. For example, if decision trees or rule-based models contain too many rules, they may become more difficult to understand. Similarly, scoring models and GAMs can become overly complex if the number of additive terms or shape functions is too large. The same is true, for example, for a linear regression model and its coefficients when the model contains hundreds or thousands of features and complex interaction terms, as the sheer number of parameters can obscure understanding (e.g., Molnar et al. 2020).

3 A Framework for Differentiating Inherently Interpretable and Post-hoc Explainable Machine Learning

In this section, we present a framework for differentiating the two paradigms of inherently interpretable and post-hoc explainable ML (see Fig. 5). The framework is divided into seven dimensions along the process of analytical model building (Janiesch et al. 2021) to highlight key principles and distinguishing aspects. In the following, we use the dimensions to discuss the main differences between the two paradigms, as well as their merits and limitations.

3.1 Areas of Application

The post-hoc explainability paradigm focuses on the development and application of methods to simplify the decision logic of ML models that are not directly understandable to human users (often referred to as black-box models). The paradigm therefore encourages the use of highly flexible and complex models such as gradient boosted decision trees or deep neural network models, which are known for their strong predictive performance. In return, post-hoc explanation methods like SHAP, LIME, partial dependence plots, class activation methods, or layer-wise relevance propagation are required to translate their complex mathematical functions into simpler, human-friendly forms (Du et al. 2019; Barredo Arrieta et al. 2020; Molnar 2025). This approach is generally suitable for application scenarios in which a high level of prediction performance is the most critical criterion for successful implementation, while model transparency is only of secondary importance (e.g., in low-risk, monetary-focused tasks like targeted advertising, churn prediction, or movie recommendations). Post-hoc methods may also be necessary if a certain level of prediction quality can only be achieved with a flexible black-box model, or if the use of such a model is mandatory and thus unavoidable (e.g., due to contractual agreements or running legacy systems). In addition, combining flexible ML models with post-hoc explanations can also be beneficial to reveal previously unknown relationships and intricate patterns that

Age Group

Factor	Coefficient
(Intercept)	-0.531
Age	+1.242

Metabolic Group

Factor	Coefficient
(Intercept)	-1.427
BMI	+0.786
HbA1c	+2.140

Meta Model

Factor	Coefficient
(Intercept)	-5.252
Age-Group	+4.549
Metabolic-Group	+5.715

Fig. 4 Subscale model comprising two subgroup models for age group and metabolic group, and a meta model. With the stacked approach, the model can be interpreted at multiple levels

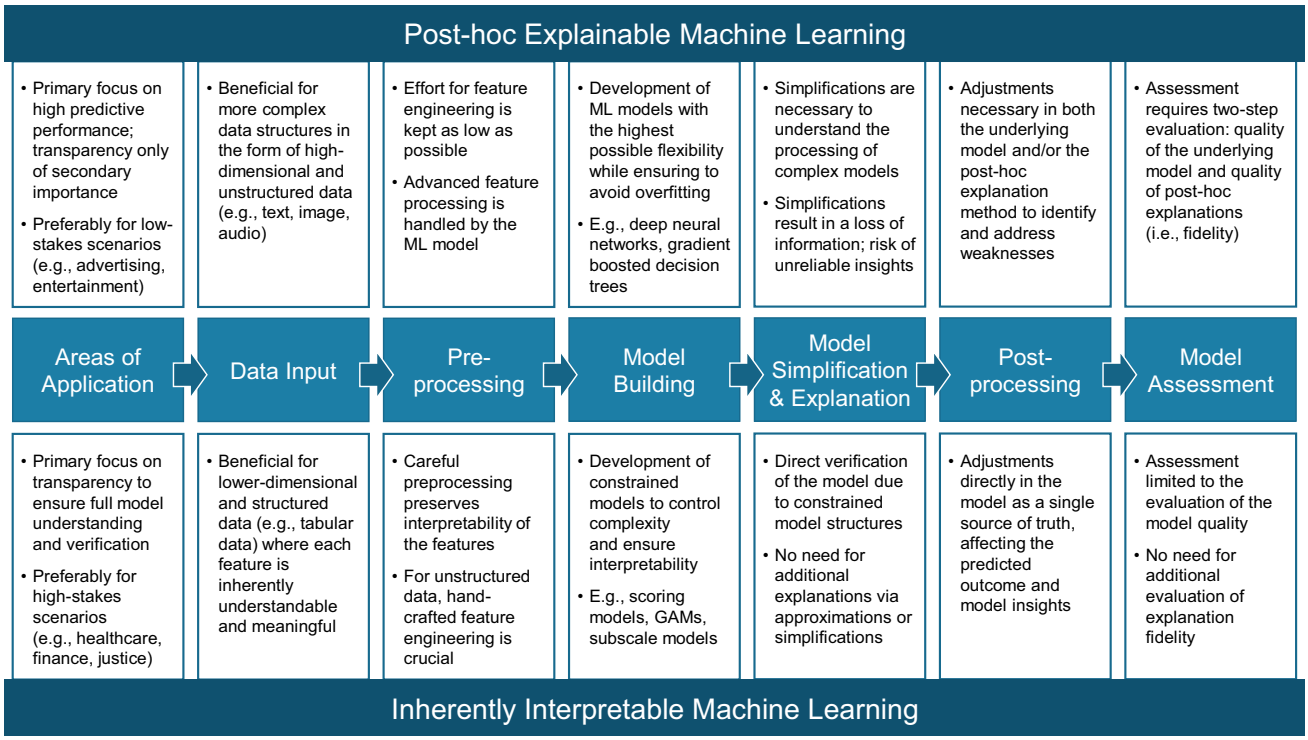


Fig. 5 Framework for differentiating inherently interpretable and post-hoc explainable machine learning

constrained ML models may not uncover (Adadi and Berrada 2018).

In contrast, the paradigm of inherent interpretability is concerned with the development and application of models that are designed to be intrinsically understandable to human users. By limiting the complexity of the models, they ensure transparency in decision making, allowing full understanding and verification of the decision logic without the need for additional explanations (Rudin 2019). This approach is relevant in high-stakes scenarios where transparency, accountability, and reliability are critical, such as healthcare diagnosis, legal decision-making, and credit scoring. For example, in medical applications, clinicians need to fully understand the model's reasoning to prevent harm and ensure patient safety, while in legal contexts, transparency is necessary to justify decisions in court. Similarly, in finance, regulatory environments often require fully transparent models to explain credit approvals or denials to ensure fairness and compliance with strict safety standards (Rudin and Radin 2019).

3.2 Data Input

To carve out the merits and limitations of both paradigms, it is important to consider the type of data input that an ML model must process. A general rule of thumb is that the more complex the underlying data and the patterns hidden within, the more flexible the ML model needs to be. For example, when working with high-dimensional unstructured data, such as text, images, and audio, it is beneficial to apply sophisticated deep neural networks such as recurrent neural networks, convolutional neural networks, or transformers (e.g., Janiesch et al. 2021). These models excel in processing high-dimensional features and transforming raw inputs like tokens or pixels into abstract, informative representations by employing automated feature extraction and representation learning techniques. For instance, in image classification, convolutional neural networks can automatically learn to identify edges, shapes, and textures in an image of a car, gradually building up layers of features that eventually allow them to classify the image correctly as a car. This enables them to capture intricate and multi-layered patterns with high predictive power that would be missed by simpler models (LeCun et al. 2015). However, because the transformed features (like pixel patterns or text embeddings) are often not directly interpretable, post-hoc methods have to be employed to provide explanations, translating these complex feature operations back into a form that humans can understand (Barredo Arrieta et al. 2020).

In contrast, inherently interpretable models excel at handling lower-dimensional, structured data, such as tabular datasets, where each feature is inherently understandable and meaningful (e.g., a person's age, income, or marital status). Because of their simplicity, these models ensure that the relationships between features and

predictions are directly traceable, allowing users to easily see how input features affect the predicted outcome. It is important to note, however, that while some interpretable models are limited to simpler data structures and may struggle with more complex patterns (e.g., linear regression struggles in the presence of non-linear relationships), there are models that can handle moderate complexity without sacrificing interpretability. Promising examples are modern variants of GAMs, such as EBM or IGANN (see Sect. 2.3). These models have shown competitive predictive performance on tabular datasets compared to more complex models such as gradient boosted decision trees, random forests, and neural networks, while remaining fully interpretable (Bohlen et al. 2025; Kruschel et al. 2025; Wang et al. 2023). For structured data with meaningful features, this is an important lesson, as it is often mistakenly assumed that only complex black-box models can achieve high predictive performance across all types of data. In fact, for tabular data, there is often no strict trade-off between the models' performance and interpretability, making interpretable models the preferred choice in many cases (Rudin 2019).

3.3 Preprocessing

By focusing on highly flexible ML models, the post-hoc explainability paradigm seeks not only to achieve high predictive performance but also to reduce the effort required for time-consuming preprocessing steps. Beyond essential data preparation tasks such as cleaning noise, removing outliers, and handling missing values, which are important in almost any data science project, flexible ML models often have the advantage of incorporating crucial feature processing operations directly within the model-building process (Janiesch et al. 2021). In deep neural networks, this capability is commonly known as deep learning or representation learning and is often a primary argument in favor of their use (LeCun et al. 2015). For example, in sentiment analysis, a deep neural network can process raw text with minimal tokenization or encoding, while deeper layers of the model extract intricate linguistic patterns that aid in classification. Conversely, sophisticated model simplification and explanation steps (see Sect. 3.6) are applied in order to gain an understanding of how the raw input data is processed by the model.

In the realm of inherently interpretable models, preprocessing plays a much more central role in the entire analytical model-building process. The aim is to ensure that the feature engineering steps support the interpretability of the model results without adding unnecessary complexity. When working with structured tabular data, this often involves careful feature selection to eliminate redundancy, reduce model complexity, and avoid issues like multicollinearity or concurvity (Bohlen et al. 2025; Siems et al. 2023). This step is crucial because even inherently simple model structures, as present in a linear regression model, can lose their practical interpretability if applied to an overwhelming number of features. Another crucial task is the construction of well-defined, straightforward features. For example, in medical applications where features are collected as time-series data, simple feature aggregations like mean or median over time windows are often preferred, while more complex metrics, such as skewness or kurtosis, may be avoided to maintain understandability (Bohlen et al. 2025). Domain expertise can help identify which features are both relevant and meaningful, potentially improving the interpretability and prediction performance of the model. Similarly, when working with unstructured data, well-defined hand-crafted features (e.g., color ratios or shape descriptors for images; frequency of key terms or parts of speech counts for text) can ensure that a model remains fully interpretable (Janiesch et al. 2021). Even though this is a laborious task, it can help to ensure that the model remains completely comprehensible and controllable.

3.4 Model Building

In the post-hoc explainability paradigm, the focus is on building black-box models that gain as much predictive performance as possible by maximizing model flexibility. Similarly, it is ensured that model overfitting is avoided, and accepted that increasing the model's performance through higher flexibility will also increase the complexity of the model. For example, in deep neural networks, high flexibility mainly results from learning multiple representations (i.e., layers) with increasing abstraction (LeCun et al. 2015), whereas, in gradient boosted decision trees, the high flexibility is largely due to their ability to iteratively build an ensemble of weak learners that correct each other's errors (Friedman 2001).

In the IIML paradigm, particular care is taken during the model-building phase to preserve interpretability. While the approach aims to maintain a high degree of flexibility, this is always balanced against the principle that complexity should not exceed a level at which humans can still understand the model. In particular, interpretability is achieved through the thoughtful integration of various constraints, which can be broadly categorized as soft constraints and hard constraints. Soft constraints include penalty terms added to a loss function and are taken into account during training. For instance, regularization techniques like LASSO (Least Absolute Shrinkage and Selection Operator) penalize model complexity by shrinking some coefficients towards or exactly to zero. This effectively performs feature selection within the model training

process, leading to sparser, potentially more interpretable linear models by reducing the number of active predictors. Hard constraints, on the other hand, are rules directly integrated into the model architecture. By integrating these constraints, the flexibility of the model is restricted, which can lead to a reduction in the models' predictive performance (Rudin 2019). Interpretable models enable the integration of additional knowledge to restrict the solution space and to guide the models' training process toward more robust solutions. For example, hard monotonic model constraints, which are previously derived from expert or domain knowledge, can be directly integrated into the decision-tree induction of a tree-based GAM to restrict the course of shape functions (Watanabe et al. 2021). On the other hand, the built interpretable models can be too constrained and not flexible enough to capture more complex relationships, such as higher-order interactions, that may exist in real-world data.

3.5 Model Simplification and Explanation

In the post-hoc explainability paradigm, black-box models typically operate in high-dimensional spaces, where complex interactions between features are common and often essential for achieving strong predictive performance. For example, deep neural networks generally operate in 100 to 10,000 dimensional spaces (the number of neurons in one hidden layer), gradually transforming them into semantically meaningful representations that enable accurate predictions (LeCun et al. 2015). This complexity makes it difficult for Humans to intuitively understand how these models reach their decisions, as we struggle to comprehend processes in such high-dimensional spaces. Consequently, post-hoc methods are necessary to simplify these processes into 1- or 2-dimensional approximations that are easier for humans to understand. Naturally, this approximation from high- to low-dimensional spaces leads to a loss of information. Post-hoc methods can simplify the complex processes of black-box models into local approximations for specific instances. For example, SHAP uses ideas from game theory to assign each feature of a given instance a certain contribution for making a particular model prediction (Lundberg et al. 2020). As another example, Diverse Counterfactual Explanation (DICE) generates counterfactual examples for instances, representing 'what if' explanations for model predictions (Mothilal et al. 2020). More recent approaches such as EXPLORE (Heinrich et al. 2024) aim to improve the fidelity and robustness of local explanations by overcoming the limitations of approximating the black-box models. Post-hoc methods can also simplify the complex processes of back-box models into global approximations of the entire model. For example, Accumulated Local Effects (ALE) describes how features influence model predictions on average (Apley and Zhu 2020). Another example is the permutation feature importance method of Molnar et al. (2024), which computes feature importance in subgroups of the data and aggregates them. However, the local and global simplifications from the post-hoc methods may fail to accurately capture the model's overall decision logic, thereby limiting its representational accuracy. Thus, it can be argued that a complex black-box model can never be fully explained in simplified terms without sacrificing crucial information, which may lead to unreliable and misleading insights (Rudin 2019).

In contrast, the paradigm of inherent interpretability enables a direct verification of the model and reduces the risk of misinterpretation, as the model's design itself is structured to produce comprehensible outputs. By constraining complexity within the model architecture, these approaches make it possible to directly observe the internal decision logic of a model, from data input to prediction. This eliminates the need for additional layers of approximation or simplification. For example, in a decision tree model, the exact rule-based trace from the root to a node leading to a prediction can be seen, allowing humans to follow the decision-making process step-by-step (Rudin 2019). Similarly, interpretable variants of GAMs allow to examine how each feature contributes to the model's predictions in a straightforward, human-understandable format (Kruschel et al. 2025).

3.6 Postprocessing

Postprocessing includes all adjustments required after reviewing the results from the model-building and explanation phases. In the post-hoc explainability paradigm, this includes tuning the explanation method and modifying the underlying black-box model, especially when issues like fairness or domain misalignment arise. A key flexibility in this paradigm is the ability to influence the nature and detail of explanations. By selecting from various model-agnostic explanation techniques or tuning their hyperparameters, practitioners can often tailor the insights provided for a given black-box model without the need for altering the underlying predictive model itself. For instance, LIME's kernel width determines the weight assigned to perturbed samples, with higher values offering a more global perspective and lower values providing more localized insights (Ribeiro et al. 2016). However, selecting optimal hyperparameters for post-hoc methods is usually challenging, as they depend on the task, data, and model complexity (Laugel et al. 2018). Moreover, when explanations remain unintuitive despite tuning the explanation method, more substantial interventions may be required. This might involve adjusting the black-box model's

hyperparameters or modifying the input data, followed by retraining. The generated explanations from the old and new models are then compared, and this iterative process may need to be repeated several times until the underlying issues are resolved.

In IIML, postprocessing focuses on adjustments to the built interpretable models, with the goal of further increasing the models' interpretability. Conversely, the adjustment of interpretability in IIML is intrinsically tied to the model's design. Significantly altering the 'explanation' usually requires modifying the model's architecture, its inherent constraints, or its learned effects. By definition, this directly impacts and changes the predictive model itself. For example, modifications of built interpretable models can include the adjustment of shape functions and centering features effects in GAMs, the adjustment of split values at nodes of decision tree models, or the integration of further regularization constraints into linear regression models. These modifications can directly address model weaknesses, such as inconsistencies with domain knowledge, misleading feature effects due to data quality issues, or potentially harmful model structures that may violate corporate requirements.

3.7 Model Assessment

In the post-hoc explainability paradigm, assessing model behavior typically requires a two-step evaluation process: one to evaluate the model's predictive performance and another to assess the reliability of the post-hoc explanations. Each step introduces its own source of potential error, which can accumulate and lead to incomplete or misleading assessments. For example, while the model might perform well on a standard test set, the explanation model might still fail to provide meaningful insights into the black-box model's behavior (Chromik et al. 2021). A key concern here is fidelity, i.e., how well the post-hoc explanations truly represent the internal decision processes of the black-box model (Kim et al. 2023). Low-fidelity explanations can lead users to trust incorrect or oversimplified insights.

In contrast, inherently interpretable models simplify the assessment process by eliminating the need for additional evaluation of explanation fidelity. This enables a comprehensive assessment, as model behavior can be evaluated for any potential input. Furthermore, it is possible to quantify aspects of interpretability directly within the model, such as properties of shape functions in GAMs, to verify that the relationships align with expected domain knowledge or desired constraints (Kraus et al. 2024). With interpretable models, humans can systematically inspect the model's decision logic across the entire input space, allowing proactive identification and correction of potential errors or biases without the need for model retraining or extensive data sampling.

4 Research Opportunities

The field of IIML offers various research opportunities for the BISE/IS community to address current challenges in research and industry and to create forward-looking innovations from an interdisciplinary, socio-technical perspective. In Table 2, we summarize some promising research directions and guiding questions for future work.

4.1 Application-Specific Definitions and Evaluations of Interpretability

In Sect. 2.1, we outlined various conceptual facets of the term 'interpretability' and emphasized the challenge of defining a universal standard from a cognitive perspective, as human interpretation processes are inherently subjective and context-dependent. As a result, algorithmic developments in IIML and XAI are often criticized for lacking clarity on whether they truly enhance human interpretation or simply operate under the broad and ambiguous label of interpretability (Broniatowski 2021; Lipton 2018; Miller 2019). Since a universally accepted definition is unlikely to emerge (Longo et al. 2024), it is all the more important to establish application-specific definitions that reflect the needs of particular domains, user groups, and tasks. These definitions should account for factors such as preferred explanation formats (e.g., rule-based, graph-based, textual), the appropriate level of detail (e.g., fine-grained vs. high-level), the degree of desired interaction (e.g., static vs. dynamic), and domain-specific conventions or cognitive expectations. While foundational algorithmic research may continue to focus on developing general-purpose methods with configurable parameters and outputs, the BISE/IS community, rooted in socio-technical thinking, is uniquely positioned to translate algorithmic innovations into usable tools by aligning them with application-specific interpretability requirements. To elicit such requirements, a variety of methods can be employed, including simulations based on expert knowledge, user-centered experiments, and hybrid methods that combine quantitative model diagnostics with qualitative user feedback. In addition to established empirical instruments like surveys and interviews, advanced techniques from the field of Neuro-IS, such as eye tracking, EEG (electroencephalography), or fMRI (functional magnetic resonance imaging), offer promising opportunities to capture implicit cognitive processes during the interpretation of ML outputs (Wang et al. 2025). These methods provide a more objective and fine-grained complement to potentially biased self-reports.

Table 2 Exemplary research directions for future research

Area	Research directions and guiding questions
Application-Specific Definitions and Evaluations of Interpretability	How can application-specific interpretability requirements be systematically identified and incorporated into the design and evaluation of interpretable ML models?
	How can Neuro-IS methods be leveraged to uncover implicit cognitive processes during model interaction to enhance the understanding of human interpretability?
	How can a unified evaluation framework be developed that combines objective model-based metrics with subjective, human-centered criteria to comprehensively assess interpretability in socio-technical contexts?
Need for Personalization in Interpretable and Explainable ML	How can interpretable and explainable ML systems adapt to the diverse cognitive styles, expertise levels, and preferences of individual users?
	How can personalized explanation systems leverage user feedback to iteratively refine and improve explanations over time?
	How do personalization requirements differ across different domains and contexts, and what implications do these differences have for explanation design?
Interacting with Explanations and Incorporating Domain Knowledge	What are effective mechanisms and interfaces for users to incorporate their domain expertise and mental models into IIML models?
	What novel ways of interaction can make IIML model explanations more accessible and actionable for users?
	How does adjusting IIML models based on their domain knowledge and insights from explanations improve model alignment with the real-world, user trust, and overall decision-making effectiveness?
Interpretability of Large Language Models	How can novel modeling languages be designed to specify the structure and the behavior of large language models?
	What concepts from process management can be used to better understand the internal mechanisms of large language models?
	How can methods, techniques, and tools be developed which explain the internal mechanisms of large language models?
Synergies between Inherent Interpretability and Post-Hoc Explainability	How can elements of IIML enhance post-hoc explainability methods and vice versa?
	How can sophisticated, interpretable models be used as local approximators within post-hoc explanation techniques?
	How can prototype learning be leveraged to develop novel explanation techniques at the intersection of IIML and post-hoc XAI?
Potential Use for Theory Development and Testing	How can interpretable models be leveraged for integrative modeling to support theory development and testing of behavioral and organizational phenomena?
	How can the principles of IIML and causal ML be combined to enhance causal inference and model transparency?
	How can new frameworks and methods be developed to quantify the uncertainty in explanations generated by interpretable models?
Importance for Ethical and Fair AI Applications	How can fairness constraints be integrated into interpretable models without compromising their transparency or exacerbating unintended biases?
	What frameworks can be developed to evaluate and balance the trade-offs between fairness, accuracy, and interpretability in ML models?
	How can interdisciplinary approaches be incorporated into the development and evaluation of interpretable models to address fairness more comprehensively?
Necessity for Regulatory Requirements	How can IIML-based AI systems or models be designed that are compliant with regulatory frameworks?
	What is the influence of IIML on the implementation of regulatory requirements by organizations providing AI systems or ML models?
	How can IIML-based approaches be developed to assess and ensure the compliance of AI systems or ML models with regulatory requirements?
Role of Design Science Research for Future Developments	How can DSR methodologies be applied to develop new interpretable models and IIML-based systems from a broader socio-technical perspective?
	What are reusable design principles of IIML-based systems?
	How can action design research contribute to the iterative refinement of IIML-based systems, ensuring their alignment with real-world challenges?

Importantly, the need for application-specific considerations extends beyond the definition phase to the evaluation of interpretability itself. While much of the current literature still relies on functionally-grounded evaluations (e.g., assessing properties like sparsity, monotonicity, or linearity), such metrics often fall short in capturing the cognitive and human-centric dimensions. Accordingly, there is a growing need for human-grounded and application-grounded evaluation methods that reflect how well explanations support real users in real tasks (Doshi-Velez and Kim 2017). At the same time, evaluations tailored to specific applications may suffer from limited standardization and high subjectivity, hindering generalizability and comparability across studies. Therefore, we advocate for the development of a unified evaluation framework that integrates objective model-based metrics with subjective, human-centered criteria. Such a framework would help balance methodological rigor with practical relevance, enabling a more robust and meaningful assessment of interpretability in diverse socio-technical contexts.

4.2 Need for Personalization in Interpretable and Explainable ML

Building upon the call for application-specific definitions and evaluations of interpretability, it becomes evident that interpretability does not only vary across domains, tasks, and user groups, it can also vary considerably between individuals. At its most granular level, interpretability can be understood as a deeply personal, user-specific concept, shaped by a person's cognitive style, expertise, goals, and context of use (Broniatowski 2021). Consequently, uniform, one-size-fits-all explanations often fail to meet the needs of diverse individuals (e.g., Hohman et al. 2019). To address this challenge, personalization in IIML and XAI aims to tailor explanations to the specific needs and preferences of individual users (Rosenberger et al. 2025). Personalized explanations are more likely to be comprehensible, actionable, and aligned with users' mental models, especially in high-stakes domains like healthcare, finance, or public administration, where trust and clarity are critical. To achieve effective personalization, we need to better understand how the inherent possibilities and constraints of interpretability differ between IIML and XAI paradigms, and how these differences affect users seeking tailored insights. For instance, XAI offers the flexibility to tailor explanations by selecting from various post-hoc techniques or tuning their parameters, without altering the core predictive model itself. Conversely, personalizing the 'explanation' in IIML is intrinsically tied to its model, meaning that adjustments to the interpretability directly impact the predictive model. Future research must explore how users perceive and interact with these distinct pathways to personalization, and how the differing mutability of interpretability influences user trust, cognitive load, and the ultimate utility of personalized explanations. Recent advances in adaptive methods illustrate the growing potential of this approach. For example, contextual bandits have been employed to dynamically adjust explanations based on user preferences (e.g., Rosenberger et al. 2025; Schröppel and Förster 2024). Likewise, large language models (LLMs) and other generative AI tools offer new ways to produce interactive, conversational explanations that can flexibly adjust to user knowledge and preferences (e.g., Martens et al. 2025; Slack et al. 2023).

Nonetheless, key challenges remain. Open questions include how to effectively model user profiles, how to adapt explanation complexity for experts versus laypersons, and how to ensure that personalization does not introduce unwanted biases or reduce transparency. Moreover, systematic methods are needed to incorporate user feedback over time, enabling explanations to evolve with users' changing expectations and decision contexts. By systematically addressing these challenges and extending the concept of interpretability from the application level to the individual level, the BISE/IS community can play a key role in advancing the design and evaluation of personalized interpretable and explainable AI.

4.3 Interacting with Explanations and Incorporating Domain Knowledge

Another research opportunity involves moving beyond unidirectional information flow in explanations (from model to user) towards systems that actively engage users in a collaborative model refinement or explanation process. This includes developing effective mechanisms for users to incorporate their rich domain expertise and mental models directly into IIML models, and designing interactive explanation interfaces that support better understanding of the model and actionable feedback. Such bidirectional interaction promises to enhance both trustworthiness and alignment with the real world.

The inherent transparency of IIML models offers a natural foundation for incorporating domain expertise. Unlike black-box models, the explicit structure of an IIML model can be directly modified by users. This aligns with the principles of *'informed machine learning'* (Von Rueden et al. 2021), where additional knowledge (beyond the training data) is used to guide the learning or explanation process (Beckh et al. 2023). For instance, domain experts, after reviewing an IIML model, might identify that a learned relationship contradicts established scientific principles. Research is needed to develop effective mechanisms and intuitive interfaces that allow users to translate their domain knowledge into model adjustments.

One promising path for enhancing user interaction with explanations, for both IIML and broader explainable ML, comes in the form of LLMs. Recent explorations have demonstrated the potential to process technical model outputs and translate them into more accessible, natural language narratives, thereby acting as a bridge to user comprehension (Martens et al. 2025; Slack et al. 2023). Beyond simple translation, LLMs could facilitate conversational interactions where users can ask follow-up questions about an IIML model's logic, request clarifications on specific components, or explore counterfactual scenarios in an intuitive dialogue (Slack et al. 2023). However, designing such interactive explanation interfaces requires careful consideration. Key research questions include how to ensure the LLM-generated explanations remain faithful to the IIML model's inherent logic, how to adapt the dialogue to a user's evolving understanding and informational needs (see also Sect. 4.2), and how to prevent the introduction of new biases or misinterpretations through the LLM.

4.4 Inherent Interpretability of Large Language Models

According to the definition of IIML (Sect. 2.1), inherently interpretable LLMs conceptually exist only to a certain extent (Luo and Specia 2024; Schneider 2024; Zhao et al. 2024). This is mainly due to fundamental differences between LLMs and inherently interpretable models: LLMs, with several millions or billions of parameters, are very flexible models that can capture very complex relationships through training on huge amounts of data, whereas inherently interpretable models integrate constraints to capture certain relationships as controlled as possible through training on curated data. Nevertheless, the development of explanation approaches for LLMs has recently received increasing attention. Some of these methods are model-specific and belong to research streams focusing on the explanation of LLM internals.

One of these research streams is *'mechanistic interpretability'* (Saphra and Wiegreffe 2024), which aims to reverse-engineer components of neural network models into human-understandable mechanisms (Bereska and Gavves 2024). Its focus was originally on the architecture of convolutional neural networks and shifted to the transformers' architecture due to the increasing importance of LLMs (Saphra and Wiegreffe 2024). Furthermore, current research can be roughly divided into a passive and an active variant (Liu et al. 2024). Most research is passive and focuses on the understanding of parts of existing neural networks using additional methods, such as circuit discovery and causal tracing (Luo and Specia 2024). In contrast, active mechanistic interpretability makes initial attempts to achieve interpretability by designing inherently interpretable neural network architectures or developing training methods that explicitly promote interpretability. However, current approaches, such as Kolmogorov-Arnold networks (Liu et al. 2024) may only work in highly simplified synthetic settings where the mathematical function mapping the data points to the label is known, but are still far from real-world applications, where external influences (e.g., noise) exist in the data (Shen et al. 2024). Nevertheless, it should be noted that basic concepts of IIML, such as sparsity, modularity, and disentanglement, are common in passive and active mechanistic interpretability, showing a clear link to IIML (Bereska and Gavves 2024). Thus, concepts of IIML may be the basis for the development of the first inherently interpretable LLMs in the future.

As a part of this development, the BISE/IS community, with its strong historical focus on structuring complex systems, can contribute by developing novel domain-specific modeling languages that allow the specification of the structure and behavior of complex LLMs. Similarly, the BISE/IS community can draw on concepts from process management like process definitions, life cycles, or modeling languages to better understand internal dependencies between neurons, information flows along different layers, or the semantic meaning of neurons depending on the time. Our community can also develop methods, techniques, and tools to explain the internal mechanisms of LLMs.

4.5 Synergies Between Inherent Interpretability and Post-hoc Explainability

The fields of IIML and post-hoc XAI have traditionally been viewed as separate, often opposing paradigms, with IIML focusing on intrinsically understandable models and post-hoc XAI on creating explanations for complex models. However, recent developments suggest a growing synergy between the two, where elements of IIML can enhance post-hoc explainability methods and vice versa. For example, integrating inherently interpretable components (e.g., concepts of GAMs) into deep learning architectures offers a promising way to build interpretable deep neural networks (e.g., Kraus et al. 2024; Yang et al. 2021; Zilker et al. 2024). This hybrid approach demonstrates that interpretability does not necessarily have to come at the cost of flexibility, enabling more transparent and user-friendly deep learning models.

Another notable area of convergence lies in the use of sophisticated, interpretable models as local approximators within post-hoc explanation techniques such as LIME (Ribeiro et al. 2016). Traditionally, LIME uses simple linear models or decision tree models as local surrogate explainers, which can fail to capture complex relationships present in the underlying black-box model. By employing

more advanced interpretable models, such as advanced rule-based models or different types of GAMs, as local approximators, the fidelity of the explanations can be significantly improved. These interpretable models, designed to capture complex patterns while remaining comprehensible, can provide richer, more accurate local explanations, making post-hoc methods more reliable.

Prototype learning, another emerging concept at the intersection of IIML and post-hoc XAI, aims to explain model predictions through the distance to learned prototypes (artificially created samples that represent, for instance, a certain prediction class) (Li et al. 2018). By learning a set of prototypes, the model provides intuitive, case-based explanations that are easily understood by human users (Rudin 2019). Thereby, the prototypes are generally learned through black-box models, mapping, for example, images into a latent, high-dimensional space. However, the explanation is then simply based on the distance between the sample to be explained and the learned prototypes. This approach highlights how the interplay between the two paradigms can foster novel explanation techniques that align well with human cognitive processes.

4.6 Potential Use for Theory Development and Testing

As emphasized in Sect. 2.1, neither post-hoc explanations nor inherently interpretable insights can directly answer causal questions, as they are based on correlational patterns. Nevertheless, ML-based explanations from both paradigms can serve as valuable starting points for causal analyses and act as complementary tools for theory development and testing in behavioral and organizational BISE/IS research (Barredo Arrieta et al. 2020). In this respect, we see an important role of such tools for *'integrative modeling'*, which seeks to complement the principles of predictive modeling (to derive data-driven models with high predictive accuracy) and explanatory modeling (to build models that reflect theory-informed causal relationships) (Hofman et al. 2021). For example, post-hoc XAI techniques, when applied to highly flexible ML models, can help exploit complex data structures and reveal potentially novel associations. This may enable researchers to identify unexpected relationships and generate new hypotheses and theories grounded in empirical evidence. Conversely, IIML models with strict interpretability constraints can ensure that learned model structures align with meaningful, theory-informed properties of real-world phenomena, offering a viable instrument to confirm or challenge existing theoretical frameworks and to refine constructs based on observable variables (Freiesleben et al. 2024). Similarly, IIML models could be used for exploratory purposes by deliberately keeping certain model components flexible, while others are constrained to control the complexity of learned relationships. For instance, advanced GAMs with a controllable number of interactions could be used to uncover non-linear relationships and identify relevant moderating effects in complex datasets. By offering a controlled balance between flexibility and interpretability, such model architectures could allow researchers to mentally simulate the underlying data-generating process and support the identification of potentially causal factors that can be systematically tested in follow-up analyses (Hofman et al. 2021).

Closely aligned with the idea of integrative modeling, exploring synergies with the emerging field of *'causal machine learning'* represents another promising avenue for future research. Unlike traditional ML, causal ML aims to identify and estimate cause-and-effect relationships to measure the impact of interventions or actions (i.e., treatments) on an outcome (Feuerriegel et al. 2024). Thus, a synergistic relationship between IIML and causal ML could benefit both fields. The field of IIML could gain deeper insights into ensuring the identification of causal mechanisms, while causal ML based on black-box models (e.g., causal forests) could be enriched by incorporating technical advancements from inherently interpretable models.

Another promising direction involves the quantification of uncertainty (Molnar et al. 2022). While numerous methods exist to quantify prediction uncertainty (e.g., conformal prediction, bootstrapping, Monte Carlo dropout, etc.), there is still a lack of well-established frameworks and methods for quantifying the uncertainty of model explanations. By advancing this area, researchers can more robustly support the theoretical insights derived from interpretable models, ultimately enabling more reliable and scientifically grounded conclusions.

4.7 Importance for Ethical and Fair AI Applications

In contrast to black-box models that require potentially flawed post-hoc explanations to diagnose their decision logic, IIML offers an ethically preferable alternative for high-stakes domains like healthcare or finance (Rudin 2019; Wang et al. 2023). Due to their transparent structures, interpretable models allow practitioners and regulators to identify and mitigate sources of bias more effectively. Thus, it is possible to evaluate how sensitive features, such as race, gender, and age, influence the model's decision logic and whether they lead to critical differences in predictions. As a result, these models can be directly adjusted to minimize the risk of harmful or discriminatory outcomes. For example, rule-based models can be modified by refining or removing rules that disproportionately disadvantage certain groups, whereas GAMs can

be adjusted by modifying the shape functions of protected features to ensure equitable treatment (e.g., Kraus et al. 2024). Another mitigation strategy could be the development of interpretable models that directly incorporate fairness constraints (e.g., demographic parity, equalized odds) into their training process. This could involve modifying training loss functions and/or regularizing feature contributions to ensure that sensitive attributes do not disproportionately influence predictions (Wang et al. 2023).

However, any kind of mitigation strategy has to be considered with caution because fairness is a multidimensional concept with many conflicting notions, making it challenging to define universally fair solutions (Deck et al. 2024). What constitutes fair treatment for one group might inadvertently lead to perceived or actual disadvantages for another. For example, fairness interventions may unintentionally perpetuate bias through proxy variables (e.g., using ZIP codes that correlate with race) or by ignoring differences in base rates (e.g., varying default risks in loan approvals due to group-specific credit scores). Such examples underscore the risk that fairness interventions, when applied without sufficient nuance, can exacerbate rather than mitigate unfairness (Deck et al. 2024; Feuerriegel et al. 2020). Moreover, conflicts between interpretability and fairness constraints further complicate the design of ethically acceptable ML models. For example, explicitly including or excluding interactions based on sensitive attributes to enforce fairness might inadvertently reinforce or undermine subgroup-specific patterns, resulting in treatments that fail to align with equitable principles. Similarly, overly simplistic models could obscure nuanced relationships that are critical to achieving fair outcomes.

To address these challenges, future advancements must be embedded within a socio-technical context. Such a context must incorporate domain expertise, normative motivations, and a nuanced understanding of the societal and historical roots of unfairness to ensure solutions align with broader ethical goals (Deck et al. 2024). Given its interdisciplinary and human-centric orientation, the BISE/IS community is well positioned to lead these efforts, advancing research at the intersection of fairness, interpretability, and accuracy, while exploring their inherent trade-offs and limitations (Feuerriegel et al. 2020).

4.8 Necessity for Regulatory Requirements

Transparency plays an essential role in regulatory frameworks, including general ones such as the European Union (EU) AI Act (European Parliament and Council 2024), as well as industry-specific ones such as Basel III (Basel Committee 2024). While the former emphasizes transparency as a general principle in AI regulation, the latter enhances global banking resilience by prioritizing the transparency of financial models in risk management. Against this background, the BISE/IS community can develop new methods and tools for the design of IIML-based AI systems and models to ensure regulatory compliance. For example, under the EU AI Act, high-risk AI systems must provide interpretable results (Sunyaev et al. 2025). Failure to comply with transparency requirements can result in severe penalties, such as fines of up to €35 million or 7% of global annual turnover, highlighting the important role of IIML in mitigating these risks.

As another promising direction, the BISE/IS community can investigate the influence of IIML on the implementation of regulatory requirements by organizations providing AI systems or ML models. These investigations can be realized through experiments that lay the focus on the opportunities and possibilities of using IIML to fulfill regulatory requirements, or on the problems and challenges associated with it. By conducting such experiments, the understanding of the role of IIML in fulfilling regulatory requirements can be improved, allowing policymakers to effectively reflect on the established regulatory frameworks and possibly adapt certain transparency-related aspects.

The BISE/IS community can also develop IIML-based approaches for understanding the compliance of AI systems or ML models with regulatory requirements. The insights gained from the development of the first XAI-based techniques in this context can be helpful in this regard (Weinzierl et al. 2024). In general, the development of such approaches is important as regulatory frameworks are often difficult to understand, and many small and medium-sized organizations do not have legal experts to review AI systems or ML models and ensure that they comply with the relevant regulatory frameworks. On the other hand, the computation of unstructured, legal textual data could be a challenge for such approaches, because the majority of IIML algorithms and models are primarily designed for the computation of structured, tabular data.

4.9 Role of Design Science Research for Future Developments

Previous research in IIML has been focused on developing new models from a primarily technical perspective (Barredo Arrieta et al. 2020; Du et al. 2019; Miller 2019). Similarly, the evaluation of interpretability often focuses on purely functional aspects (Doshi-Velez and Kim 2017). Although these perspectives are essential, the BISE/IS community, with its strong focus on design science research (DSR) and its roots in socio-technical problem-solving, can significantly contribute to fostering more holistic developments in the future by incorporating interdisciplinary perspectives (e.g., psychology, physics,

sociology). Promising avenues are the consideration of kernel theories from cognitive sciences to inform the selection of supportive model constraints, the incorporation of domain knowledge (e.g., expert-driven, physics-informed) to guide the design of effective model structures and outputs, or the evaluation of organizational and human-centric aspects (e.g., trust, usability, willingness to use, task-technology-fit, etc.) to ensure the effectiveness of new model developments.

However, a DSR perspective offers more than just a lens for examining design aspects at a pure model level. Instead, it is particularly valuable for embedding interpretable models in broader information systems, thereby ensuring system-level usability and explainability (Bauer et al. 2021). Critical aspects at a system level include designing graphical interfaces for diverse stakeholders (e.g., developers, managers, customers), enabling different forms of user interaction (e.g., IIML model configurability, personalized views), supporting the possibility for continuous model improvement (e.g., due to concept drifts or evolving requirements), and ensuring the explainability of further pipeline components (e.g., semantic explanation of raw data). Such broader design aspects are crucial for making IIML practically applicable and valuable in addressing real-world challenges.

Initial studies have already explored the design of XAI- and IIML-based information systems using DSR methods (e.g., Kim et al. 2023; Pumplun et al. 2023). However, further research is required to assess the generalizability of derived design knowledge across different contexts, such as different domains or data types. Moreover, much of the current research focuses predominantly on post-hoc explainability approaches, highlighting the need to investigate the extent to which specific design principles are required for artifacts employing inherently interpretable models.

A particularly promising direction lies in the application of action design research initiatives (e.g., Pfeuffer et al. 2023). Action design research fosters iterative development through cycles of implementation and reflection in real-world contexts, ensuring alignment with practical constraints and stakeholder needs. Its collaborative nature can promote the identification of context-specific design principles for IIML systems and facilitates the development of solutions that are both adaptable and grounded in real-world requirements.

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